

months, and in some cases a chronic recurring tightness may restrict full athletic function.

Muscle strains occur commonly in human athletes, so it is likely that they are underdiagnosed in canine athletes.⁵⁵ It can be difficult to diagnose a muscle strain, especially because dogs with muscle strains often have concurrent orthopedic problems. Diagnosis may be made by eliciting pain when the affected myotendinous unit is directly palpated or passively moved into a stretched position. Imaging techniques such as diagnostic ultrasound or magnetic resonance imaging may be necessary to establish a diagnosis.

Management of acute muscle injuries varies with the severity of injury. Large areas of muscle disruption or total muscle rupture should be surgically repaired.⁵⁶ Grade 1 or 2 injuries are typically treated conservatively. This includes the use of analgesic agents, cryotherapy, and rest or controlled activity. Stretching is typically avoided during the acute phase of healing. Within the first week after injury or surgical repair, pulsed therapeutic ultrasound and laser may be used for their nonthermal effects on tissue repair.³ If muscle is surgically repaired, the area may be immobilized by splinting the limb for several weeks. During this time, PROM of adjacent joints is indicated. After the muscle has healed, muscle lengthening may be promoted by the use of passive stretching and active ROM exercise. Additionally, exercises may be chosen to strengthen the affected muscles. However, there are no established treatment guidelines for muscle strains in dogs and the effectiveness of physical rehabilitation has not been established.⁵⁵

Almost any muscle can sustain injury. Common muscles injured in racing greyhounds include the gracilis, tensor fascia lata, pectorals, and triceps.⁵⁶ Gracilis muscle rupture, also known as “dropped thigh” in racing greyhounds, occurs from a single traumatic event and dogs often recover after surgical repair.²⁹ The iliopsoas, pectineus, adductor, and gracilis have been reported to be affected by muscle strain in the hindlimb of dogs.⁵⁵

Eccentric muscle contraction may play an important role in the development of acute muscle strain injury because a weightbearing force exceeding the force of muscle contraction is applied to the muscle, resulting in muscle lengthening while it is contracting against the load. This leads to muscle damage. Inadequate warm-up, fatigue, or overtraining may also be factors involved in muscle strains.²⁹ In addition, after a muscle has been injured it may be at risk for reinjury, particularly if the previous muscle strain has not completely healed. Light activity is recommended to increase blood supply and muscle temperature before vigorous exercise to reduce the risk of injury. Stretching may also be performed before exercise, but stretching without an appropriate warm-up may be detrimental.

Fibrotic Myopathy of the Thigh Muscles

Fibrotic myopathy is a condition in which the muscle fibers are replaced with fibrous connective tissue, resulting in a

dense fibrous band within the muscle. It has been reported in the semimembranosus, semitendinosus, biceps femoris, gracilis, and sartorius muscles in the dog. It has been reported most commonly in German shepherds, but is also seen in Doberman pinschers, Rottweilers, and a number of other breeds.^{29,57} The etiology is unknown, but muscle trauma is suspected in some cases. The onset is insidious in some cases, leading to the speculation of chronic or recurrent muscle strain as a cause. The most common clinical sign is nonpainful lameness, although pain may be elicited on palpation of the muscles. The gait is characterized by a shortened stride with internal rotation of the stifle during the swing phase, and the stifle joint may have decreased extension. The condition can be bilateral. A fibrous band may be palpated in the affected muscle.

Surgical excision of the affected tissue may cause temporary resolution of signs, but the fibrosis typically recurs within a few months if intensive rehabilitation for several months is not performed. Rehabilitation with ultrasound, PROM, cross-fiber friction massage, and controlled exercise for several months has resulted in improvement and acceptable clinical function.²⁹ If contracture begins to develop despite rehabilitation, a dynamic splint may be used to apply chronic, passive stretch to the tissues. Maintaining tissue tension for 8 to 12 hours per day may help delay the progress of contracture until the fibrous tissue matures in an elongated position.^{3,29} Long-term monitoring and aggressive intervention may improve the chance of a successful outcome, but the prognosis is generally considered to be guarded.

Stifle Conditions

Articular Fractures of the Stifle

Fractures of the femoral condyles occur relatively rarely. Fractures of the patella may occur as a result of falling on the stifle on a hard surface, or as avulsion fractures of the quadriceps muscle. Physeal fractures of the proximal tibial physis sometimes occur. Salter-Harris Type III and IV fractures are more serious because the fracture line involves the articular surface. As with any articular fracture anatomic reduction, rigid fixation with screws and early, gentle, diligent ROM and joint mobilization are necessary for success. Surgical fixation of articular fractures must be strong enough to withstand early PROM exercises (Figure 32-27). Rehabilitation of fractures of the stifle is similar to that of fractures of the elbow (see Table 32-1).

Osteochondritis Dissecans of the Stifle

The presence of OCD in the stifle is uncommon. The lesions generally occur on the axial aspect of the lateral femoral condyle. It is usually treated by arthrotomy or arthroscopy to remove the osteochondral fragment(s), and to curette, forage, or micropick the subchondral bone



Figure 32-27 Postoperative view of an articular fracture of the distal femur. Consultation with the surgeon is recommended to determine the type and amount of exercise that can be safely done without jeopardizing the stability of the repair.

to create an opportunity for vascular ingrowth and fibrocartilage formation. Alternatively, osteochondral grafting may be an option. Nonsteroidal antiinflammatory drugs, cryotherapy, gentle PROM, and leash walks are advocated for the first 2 to 4 weeks, depending on the severity of the lesion. After that if the lameness is improving, leash walking is increased and more aggressive activities are added after 4 to 6 weeks (see [Table 32-2](#)).

Cranial Cruciate Ligament Rupture

Cranial cruciate ligament (CCL) rupture is one of the most common orthopedic conditions in dogs. Trauma-induced acute rupture of the CCL results in gross instability of the stifle, and is often accompanied by a sudden non-weight-bearing lameness, which improves gradually over the first few weeks. However, most cases of CCL rupture occur as a result of ligament degeneration. The animal may present with acute lameness or may have a more insidious lameness as partial rupture of the ligament occurs over time. In either scenario osteoarthritic changes progress with time and often result in diminished limb function.

Cranial cruciate ligament rupture is diagnosed by palpating cranial drawer motion (cranial subluxation of the tibia). Joint effusion may be evident and is best palpated adjacent to the patellar tendon. In more chronic cases drawer motion may be more difficult to elicit and the

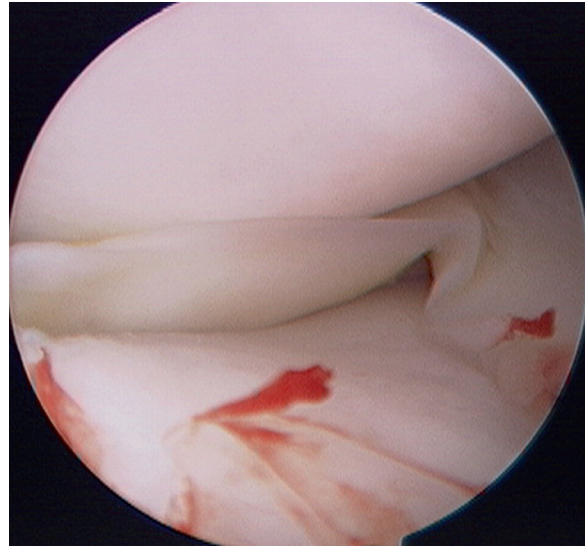


Figure 32-28 Arthroscopic view of the torn medial meniscus as it is shifted cranially within the joint. The caudal aspect of the medial meniscus is commonly damaged because of stifle instability as a result of cranial cruciate ligament rupture.

periarticular tissues may be thickened, particularly on the medial aspect of the stifle. This is termed *medial buttress*. Radiographs may be used to confirm the presence of joint effusion, assess the severity of any degenerative changes, and rule out other problems such as neoplasia.

Damage to the caudal pole of the medial meniscus often occurs as a result of joint instability ([Figure 32-28](#)). The lateral meniscus can also be damaged, but it is typically not as severely affected as the medial.⁵⁸ Damage to the meniscus rarely occurs as an isolated injury, but may be seen in approximately 45% of animals with CCL rupture.⁵⁹ In some cases the meniscus may be normal at the time of the initial surgery for CCL rupture, but become damaged at some time in the future. Clinical signs of a meniscal tear may include lameness and an audible click during joint motion. Partial or total meniscectomy via arthrotomy or arthroscopy is used to treat meniscal tear. If a meniscectomy is required the prognosis is generally somewhat worse than for stifles that have an intact meniscus. The rehabilitation program after meniscectomy is dictated by the procedure used to stabilize the joint. In some cases the meniscus is normal at the time of stifle stabilization but is damaged months or years later. This is probably the most common reason for an animal to develop acute lameness after CCL surgery.

Conservative treatment of CCL rupture by strict confinement for 4 to 8 weeks may be attempted, but the progression of OA is greater. An external brace may be used to support the stifle, but there are no studies to document their efficacy in dogs. Knee braces similar to those used in people after cruciate ligament repair may provide support and improve proprioception and limb function, but they do

Table 32-4 Rehabilitation After Surgical Repair of Ruptured Cranial Cruciate Ligament

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Maintain range of motion and joint health	Passive range of motion and/or joint mobilizations for 3 to 4 weeks. Consider aquatic walking. Add cavaletti walking after several weeks, if repair is stable enough.
Control pain	Nonsteroidal antiinflammatories $\pm$ other analgesics $\pm$ transcutaneous electrical nerve stimulation within first postoperative week. Decrease pain management as indicated by assessment of the dog.
Strengthen muscle	Consider neuromuscular electrical stimulation in early postoperative time to limit atrophy. Begin with controlled slow leash walking. Consider aquatic walking after incision has sealed. Progress to stair climbing and sit-to-stand exercises. Rate of progression is typically more conservative following tibial plateau leveling osteotomy as compared with an extracapsular suture. No explosive activity for 3 months.

not provide adequate stability and it is not clear that outcomes are improved.⁶⁰⁻⁶³ Surgery is accepted to be the treatment of choice, regardless of dog size. Many surgical techniques have been described to treat cranial cruciate ligament rupture, but none have been proven to be superior.^{65,66} Physical rehabilitation may be at least as important as surgical technique in affecting functional outcome.⁶⁷⁻⁶⁹

The prognosis for CCL rupture is variable. In humans a trend toward more aggressive rehabilitation has shown improved outcomes following surgery for cruciate ruptures.⁷⁰ Postoperative rehabilitation programs for CCL surgery in dogs improves joint ROM, reduces muscle spasms, and improves weight-bearing and overall joint function.^{67,69} Dogs treated only with NMES after CCL surgery had improved lameness scores, increased thigh circumference, and decreased radiographic changes as compared with untreated dogs.⁷¹ The focus of rehabilitation in most cases is on improving stifle extension and increasing mass of the quadriceps, biceps femoris, and semimembranosus muscles.⁷² Some dogs recover with near normal function. However, some have moderate to severe OA at the time of diagnosis, requiring long-term management (see section on OA). Total stifle replacement surgery is available, but is not commonly performed.⁷³

To simplify discussion of CCL rehabilitation, the surgical repairs are grouped into three categories. The surgical procedures have been traditionally classified as extracapsular or intracapsular, based the location of the implant or graft that is used to provide joint stability. A third approach is to change the joint biomechanics by performing some type of tibial osteotomy.

Extracapsular Stabilization

Extracapsular procedures stabilize the joint by transposition of the animal's own periarticular tissues or by securing a synthetic material (usually suture) external to the joint

capsule. Most commonly, the suture is passed around the lateral (and sometimes medial) fabella and through a tunnel drilled into the proximal tibial crest. Modifications of this procedure involve stabilizing the suture with bone anchors or through bone tunnels. These procedures provide immediate extracapsular stabilization of the stifle, allowing early rehabilitation to be initiated. At some point, the prosthetic material may fatigue and break. The joint is stabilized by periarticular fibrosis within 8 to 10 weeks, so if suture breakdown occurs several months postoperatively joint stability is not significantly affected. Nevertheless, it is common for some degree of cranial drawer motion to be present weeks to months after surgery. The complication rate associated with this procedure is relatively low and most are incisional complications.⁷⁴

Fibular head transposition is another type of extracapsular stabilization. The fibular head is transposed cranially and stabilized with a pin and tension band. The distal portion of the lateral collateral ligament inserts on the fibular head, so shifting it cranially causes it to function similar to a CCL, by limiting cranial drawer and internal rotation of the tibia. The repair provides immediate stability of the stifle joint, but some drawer motion may return with time.

Physical rehabilitation of extracapsular techniques begins in the immediate postoperative period with cryotherapy, NSAIDs, and PROM to reduce inflammation and pain (Table 32-4). The common practice of placing a compression bandage immediately postoperatively may do little to limit swelling of the joint.⁷⁵ Cold compression or microcurrent electrical therapy in combination with a compressive bandage may be effective in limiting postoperative edema.^{76,77} Electrical stimulation (TENS) may help with pain management.⁷⁸ After loss of the cruciate, even with immediate surgery, the affected leg loses muscle mass within 2 weeks and bone density within 4 weeks.^{69,79} It is

also important to establish normal ROM within 2 weeks to regain full stifle extension.⁸⁰ Physical rehabilitation can facilitate an increase in muscle mass after surgery.^{69,72,80} Neuromuscular electrical stimulation (NMES) may also be initiated to limit muscle atrophy.⁷¹ Controlled leash walks and active use of the limb are encouraged the day after surgery. Treadmill walking can be used to encourage weight bearing. Aquatic therapy can begin within 1 week postoperatively if the incision is sealed and has no drainage or discharge. Swimming or water walking may improve ROM, particularly stifle flexion.^{38,81} Aquatic therapy, whether swimming or walking, also reduces the axial load on the joint,³⁹ thus potentially limiting stress on the surgical repair. Some animals may flail their limbs in water, so aquatic therapy could be counterproductive in those cases. Passive range of motion is continued with an emphasis on stifle extension. After the inflammation has subsided, topical hot packs or continuous ultrasound may be used to limit pain, increase blood flow, and increase tissue extensibility before the rehabilitation session. Cryotherapy may be performed at the end of each exercise session to limit inflammation and pain. Hindlimb muscles can be strengthened by stair climbing, incline or decline walking, sit-to-stand exercises, pulling a cart, or dancing. As strength and endurance improve, the duration and intensity of the exercises can increase to include jogging, controlled ball playing, and swimming. However, explosive activity such as jumping is not allowed for the first 3 months.

Intracapsular Stabilization

Intracapsular procedures replace the cranial cruciate ligament in a nearly anatomic position with a graft, synthetic material, or a combination of these materials. Biologic grafts are favored over synthetic prosthetics because the latter tend to degenerate and cause chronic intraarticular inflammation. Biologic grafts undergo a period of weakness until the tissue revascularizes and gains additional strength. Months may be required for the graft to regain sufficient strength to function as a normal CCL. When creating a therapeutic plan, the therapist must consider the material used for the surgery. Autografts and allografts are generally strong when initially placed, but stretch under tension. The tissue becomes weaker over the next 2 to 20 weeks while revascularization and incorporation occur. Ultimate tissue strength is achieved following biointegration of the graft. In general, the rehabilitation program is similar to, but much more gradual than, extracapsular repair, and should not progress as rapidly because of the limited graft strength.

Tibial Osteotomies

Several tibial osteotomy techniques have been described, but the most commonly used is the tibial plateau leveling osteotomy (TPLO). Other techniques include the tibial tuberosity advancement (TTA), closing wedge tibial

osteotomy (CWTO), proximal tibial intraarticular osteotomy (PTIO), chevron wedge osteotomy, and triple tibial osteotomy (TTO). No one technique has been shown to be superior.⁸² A variety of complications have been reported for TPLO and TTA procedures, but the overall complication rates are similar.⁸³ The theory behind the osteotomy techniques is to alter the joint biomechanics to eliminate abnormal cranial tibial thrust motion during weight-bearing activity. The TPLO procedure is performed by making an osteotomy in the proximal tibia, rotating the proximal tibia to level the weight bearing surface and change the joint biomechanics, and stabilizing it with a bone plate until the bone has healed (Figure 32-29). The TTA procedure is performed by making an osteotomy in the cranial aspect of the proximal tibia, allowing the tibial tuberosity to be transposed cranially and stabilized with a special wedge-shaped implant, thus shifting the alignment of the patellar tendon and eliminating cranial tibial thrust.

The most common complication following TPLO is implant-associated infection.⁸⁴ Patellar tendinitis is a common complication that is seen in the first postoperative month, and is identified by pain on palpation of the patellar tendon at its insertion on the tibial crest. The tendon is also be palpably thickened (Figure 32-30). Another complication is avulsion of a portion of the tibial crest. Fracture of the tibial crest is also a reported complication of TTA. Although dogs may become very functional after TPLO surgery, the limb may never be completely normal.⁸⁵

The therapy of tibial osteotomy techniques is initially the same as for extracapsular techniques, and involves cryotherapy, NSAIDs, and PROM in the immediate postoperative period. Limited activity in the form of progressively longer leash walks, are begun in the initial postoperative period. Potential complications of patellar tendinopathy and tibial crest avulsion must be considered when developing the rehabilitation protocol. Excessive stress on the patellar tendon should be avoided in the early postoperative period until the tissues have a chance to adapt to the altered joint biomechanics. The quadriceps muscle-patellar tendon unit should be maintained in a relatively shortened position to reduce tensile forces on the tibial crest. Excessive flexion of the stifle during weight bearing should be avoided. This includes jumping, running, stair climbing, or walking in a crouched position. The therapist must carefully monitor the patellar tendon and tibial crest, and alter the treatment protocol if necessary. Early detection and treatment of complications should result in an earlier return to function and an improved outcome as compared with allowing the conditions to progress to the point of clinical lameness and pain. These complications usually can be managed by NSAIDs, cryotherapy, extracorporeal shock wave therapy, and more restricted activity. Use of a hinged knee brace to limit flexion of the stifle may be considered, although its efficacy has not been reported. After the tissues have healed,

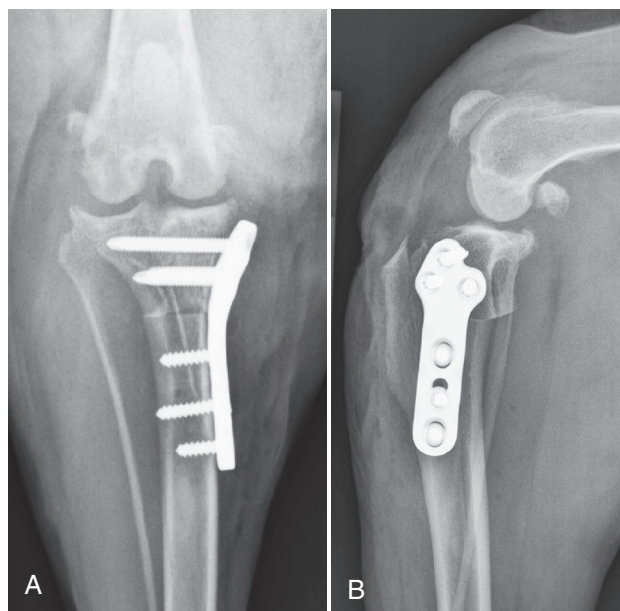


Figure 32-29 A dog with a cranial cruciate ligament injury has had the stifle stabilized with a tibial plateau leveling osteotomy. A, A-P view, B, Lateral view.



Figure 32-30 This dog became more lame several weeks after TPLO surgery. The increased soft tissue density in the cranial aspect of the joint is consistent with patellar tendonitis.

activity level can be gradually increased. The osteotomy site must be given adequate time to heal (6-10 weeks) to prevent complications related to bone healing and implant failure. As healing of the osteotomy site progresses, a gradually increasing protocol of limb use is prescribed.

Aquatic therapy may be useful to reduce weight-bearing stresses on the repair.³⁹

Caudal Cruciate Ligament Rupture

Isolated rupture of the caudal cruciate ligament is rare. The caudal cruciate ligament may rupture as a complication of TPLO overrotation, but it is usually caused by severe stifle trauma that results in stifle luxation. Treatment of isolated caudal cruciate ligament rupture is surgical stabilization with an extracapsular suture, and rehabilitation is carried out as for extracapsular stabilization of cranial cruciate ligament rupture. Treatment of caudal cruciate ligament ruptures that occurs in combination with damage to other stifle structures is discussed in the section on stifle luxation.

Stifle Luxation

Total derangement of the stifle, with rupture of both cruciate ligaments and one or both collateral ligaments, is a result of severe trauma. The menisci are often damaged as well. Collateral ligament trauma is usually best treated by surgical repair. The repair may involve suturing stretched or torn ligaments, or the use of prosthetic material (e.g., suture and screws or bone anchors) to replace the torn ligament(s). The cruciate ligaments are generally repaired with an extracapsular technique. If the damage was severe or injury was sustained to other limbs, there will be increased stress on the repaired stifle. The repair may be protected with a bandage or splint for the first 2 to 4 weeks. External coaptation and transarticular external fixation will cause at least a temporary decrease in ROM, but may be necessary.^{86,87} Hinged external fixators allow joint flexion and extension. It is not clear whether the benefits of external fixation outweigh the potential complications. If the stifle is not immobilized, immediate PROM helps to prevent joint contracture, promote cartilage homeostasis, and stimulate scar formation along normal lines of stress, resulting in a stronger scar. If the stifle is immobilized postoperatively, the goal after bandage or brace removal is to eliminate joint stiffness and re-establish ROM by implementing passive and active ROM exercises.⁸⁷ Therapeutic ultrasound with simultaneous stretching also may be beneficial. Weight-bearing activities are encouraged with weight shifting, slow leash walks, and treadmill walking. Aquatic therapy may also be helpful. Appropriate challenges to the healing tissues enhance tissue remodeling and strengthening. Endurance and strengthening activities may be initiated between 4 and 6 weeks. Near full return to activities should be achieved by 12 to 16 weeks.

The prognosis for stifle luxation is fair. Long-term problems include decreased ROM, chronic instability, and ongoing OA. Arthrodesis is an option for a chronically unstable stifle joint, but function is poor. Activity should be restricted until there is radiographic evidence of fusion, which is usually in about 8 weeks. Increased

weight-bearing exercises and muscle reconditioning are begun after the stifle has fused (see [General Physical Rehabilitation Management of Joint Arthrodesis](#)).

Patella Luxation

Although patella luxations may be traumatic in origin, they are most commonly associated with torsional deformities and malalignment of the femur and tibia. Medial luxations are more common than lateral luxations in all dog breeds and in cats.⁸⁸ Concurrent cranial cruciate ligament rupture is present in 15% to 20% of middle-aged and older dogs with chronic patella luxation.⁴⁰ Patella luxations are classified from grade I to grade IV, with IV being the most severe. Grade I patella luxation indicates that the patella stays in the trochlear groove but can be manually luxated, whereas grade IV indicates a patella that is always luxated and cannot be reduced. The higher grades usually correlate with more severe clinical signs and severity of bony abnormalities.

Surgery is only recommended when the patella luxation causes gait abnormalities or lameness; therefore surgery is generally not performed for grade I luxations. Surgery is only recommended for grade II luxations if clinical signs are causing limb dysfunction, but is usually indicated for grade III and IV luxations. Surgical correction of patella luxations usually requires reconstruction of both soft tissues and bone. In almost all cases the tibial crest is transposed and pinned to realign the quadriceps mechanism ([Figure 32-31](#)). Some type of trochleoplasty to deepen the trochlear groove is performed if the trochlear groove is shallow. A femoral osteotomy may be performed if severe varus deformity of the femur is present. The most common complications of patella luxation repair are patella relaxation and implant loosening or failure.⁸⁹ Other reported complications include trochlear wedge migration, wound dehiscence, tibial tuberosity fracture, and septic arthritis.

After surgery the tissues must be allowed to heal before vigorous rehabilitation begins, particularly if the tibial crest was transposed.

During the first few weeks cryotherapy and NSAIDs may be used for inflammation ([Table 32-5](#)). Active use of the limb is encouraged, but initial activity should be limited to short leash walks, and no running, jumping, or playing are allowed. Passive ROM is beneficial for joint resurfacing and cartilage healing, especially if a trochleoplasty was performed.⁹⁰ Small breed dogs in particular are sometimes reluctant to bear weight on the affected limb, even in the absence of apparent pain or complications. In these cases weight-shifting activities or a syringe cap placed under a

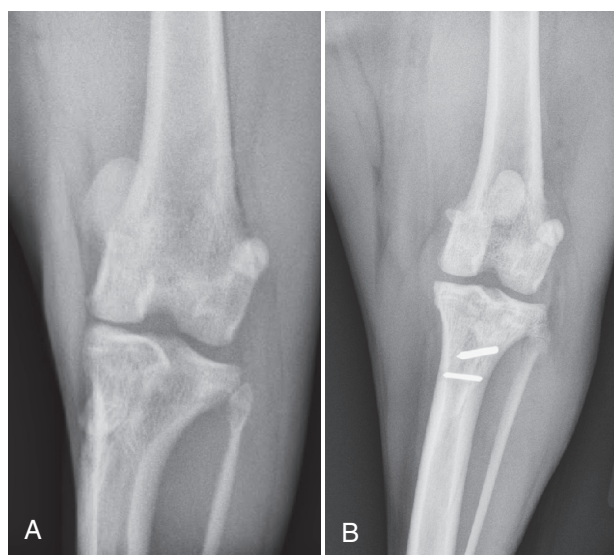


Figure 32-31 Medial luxation of the patella. (A) is the preoperative radiograph and (B) is the postoperative radiograph with pins present in the transposed tibial crest.

Table 32-5 Rehabilitation After Surgical Repair of Patella Luxation

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Maintain range of motion and joint health	Passive range of motion for 3 to 4 weeks. Consider aquatic walking.
Control pain	Nonsteroidal antiinflammatories $\pm$ other analgesics as needed after surgery. Decrease pain management as indicated by assessment of the dog.
Strengthen muscle	Initially, activities such as weight shifting to encourage weight bearing. Start with controlled leash walks. Limit activities that may put rotational stress on the stifle. Motion should be confined to the sagittal plane. After 2-3 weeks, focus on increased strength of hindlimb with walking, incline walking, sit-to-stands, dancing, etc.

normal foot may be instituted to encourage limb use. Walking on an underwater treadmill may encourage strengthening of the quadriceps muscles while maintaining the motion in a sagittal plane. After several weeks strengthening exercises may be initiated. Motion should be limited to the sagittal plane to avoid undue stress on the repair. The prognosis is generally fair to good for grades II and III, but guarded for grade IV luxations.

Rupture of the Patellar Tendon

Rupture of the patellar tendon is rare, but can occur with direct trauma or with forceful quadriceps contraction during forced flexion of the stifle. Patellar tendon rupture is treated by surgical tendon repair. A transarticular external fixator or splint may be placed for 2 to 3 weeks in an effort to limit tension on the repaired tendon. However, this will not prevent tension created by contraction of the quadriceps muscles. Superficial cold therapy is used in the postoperative period to reduce local inflammation. Pulsed 3-MHz therapeutic ultrasound applied directly over the tendon is appropriate to help stimulate collagen production. Therapeutic laser may also be beneficial. Therapy may include PROM and light activity such as leash walking to maintain normal tissue length and encourage normal joint movement. Standing and balancing exercises can promote proprioceptive input. However, active stifle motion should be limited to prevent disruption of the weakened tissue. Between 3 and 6 weeks controlled exercise should be initiated. The tendon gains significant strength by 6 weeks, but will not reach maximal strength for a year or more.⁹¹ During the remodeling phase therapy is directed at maintaining tissue mobility, preventing adhesions, and improving proprioception and muscle strength.

Traumatic Injuries of the Long Digital Extensor Tendon

Avulsion of the tendon of the long digital extensor muscle is a rare condition that occurs in young dogs. Clinical signs include joint effusion and pain in the cranio-lateral aspect of the joint. If not treated OA and chronic low-grade lameness may result. Treatment involves reattachment of the avulsed fragment of bone, or removing the fragment and attaching the tendon to the proximal tibia or joint capsule. If the bony fragment is reattached, time is allowed for union before starting aggressive rehabilitation. In the initial postoperative period, pain and inflammation are treated with NSAIDs and cryotherapy. Limited weight bearing with muscle re-education and proprioceptive training are instituted as the fracture heals. If the fragment was excised and the tendon reattached, weight bearing, controlled activity can be started soon after surgery. The prognosis is good if surgical treatment is performed before OA is apparent.⁹²

Crural Conditions

Avulsion of the Tibial Tuberosity

This condition most often occurs in immature dogs before the crest fuses with the main shaft of the tibia. Avulsion results in proximal and cranial displacement of distal portion of the tibial crest, or of the entire crest. This results in patella alta and the dog is unable to fully extend the stifle. This fracture is best treated with a pin and tension band technique to reattach the bony fragment, although minimally displaced fractures may be treated conservatively with rest and a lateral splint to limit stifle flexion. Cryotherapy, NSAIDs, and limited weight bearing are initiated after surgery. Follow-up radiographs are made 3 to 4 weeks after surgery to monitor the progress of fracture healing and guide the initiation of a more aggressive weight-bearing and muscle reeducation program.

Tibial and Fibular Fractures

Fractures of the tibia and fibula are common in the dog. Depending on the fracture type and configuration, they may be treated with closed reduction and external coaptation, or open reduction with internal or external fixation. Nonsteroidal antiinflammatory drugs, cryotherapy, and PROM are used to decrease pain and inflammation and maintain ROM for the first 2 to 3 weeks. Massage may be performed initially to help resolve edema. Active weight-bearing exercises are then initiated. The rate and progression of activity are dictated by the severity of the fracture and stability of repair. Because relatively little soft tissue covers the medial surface of the tibia, open fractures are quite common. In these cases, wound management and bandaging also may be necessary. Osteomyelitis tends to occur in the tibia more commonly than other bones and is probably related to the high incidence of open fractures. Aquatic therapy is contraindicated in patients with open wounds.

Tarsal, Metatarsal, and Phalangeal Conditions

Tarsal, Metatarsal, and Phalangeal Fractures

Fractures of the distal fibula, tibia, or tarsal bones may occur as a result of trauma, such as athletic injury or automobile accidents. Repair of medial or lateral fractures of the malleolus are important because these serve as the origins of the collateral ligaments. Metatarsal and phalangeal fractures usually occur as a result of trauma, and phalangeal fractures are often open fractures. Tarsal, metatarsal, and phalangeal fractures may be treated using closed reduction and immobilization in a coaptation device, or with a variety of implants, including intramedullary pins, wires, and plates and screws. A splint or light cast is usually

applied to the limb for several weeks after internal fixation to protect the relatively small implants that must be used.

As is the case for all articular fractures, it is important to achieve anatomic reduction and rigid fixation with screws, pins, or small bone plates. Also, early ROM is important following surgery. If a cast or splint is used, stifle ROM can be preserved while the external coaptation is in place. If possible, the cast or splint should be removed on a weekly basis to enable PROM exercises to be performed to maintain joint and tendon mobility. After performing PROM exercises, the coaptation is reapplied to protect the fracture. Following permanent removal of the splint or cast, remobilization of the joints is important, along with weight-bearing therapeutic exercises and muscle reconditioning. Sagittal-plane ROM exercises and joint mobilization work well to restore ROM to the area.

Osteochondritis Dissecans of the Tarsus

Tarsal OCD occurs less commonly than OCD in the shoulder. Lesions can occur on the medial or lateral ridge of the talus, and can be difficult to see radiographically and may be more readily diagnosed by CT or arthroscopy. The pathophysiology is the same as for OCD in other joints. Dogs with hock OCD may present with signs of OA such as joint effusion, thickening of periarticular soft tissues, decreased ROM, and crepitus. It is usually treated by arthrotomy or arthroscopy to remove the osteochondral fragment(s), and curette, forage, or micropick of the subchondral bone to allow vascular ingrowth and fibrocartilage formation. Large cartilage flaps in the hock are sometimes fixed in place in an effort to preserve joint congruency, but OA will progress. Nonsteroidal antiinflammatory drugs, cryotherapy, gentle PROM, and leash walks are advocated for the first 2 to 4 weeks, depending on the severity of the lesion. After that if the lameness is improving, leash walking is increased and more aggressive activities are added after 4 to 6 weeks. The prognosis for hock OCD is guarded because OA progresses despite treatment (see [Table 32-2](#)). In severe cases arthrodesis may be performed.

Tarsal Luxation

Most luxations of the tarsocrural joint occur with concomitant avulsion of one or both malleoli, which is where the collateral ligaments originate. Subluxation can occur with rupture or avulsion of the collateral ligaments. Traumatic shearing injuries commonly result in loss of skin, subcutaneous tissues, tendons, ligament, and bone from the medial or lateral aspect of the joint. Depending on the type of damage, surgical repair of hock luxation or subluxation may include imbricating, suturing, reattaching the ligaments, or replacing the ligaments with a prosthetic collateral ligament technique. In some cases an external fixator may be applied as the primary repair.⁹³ Open wounds are often present, which must be dressed or surgically repaired. If there is significant skin loss, a graft may

be required after the wound is healthy. Ligament repairs are typically protected during healing by rigid immobilization in a cast, splint, or external fixator for at least 3 weeks. This prevents the sutures from tearing out and prevents elongation of the ligamentous fibers. Ligament repairs are generally not able to withstand full weight-bearing stresses for several weeks. However, in joints rigidly immobilized for 6 weeks, the ligaments fail to gain as much strength as ligaments of joints that are not immobilized beyond 3 weeks.⁹¹ A hinged transarticular external fixation will allow some joint motion while protecting the primary repair, but it is not known whether the long-term outcome is improved over joints that are rigidly immobilized.⁸⁷ After the cast or external fixator is removed, remobilization of adjacent joints, weight-bearing exercises, and muscle reconditioning are begun. Active or passive therapeutic exercises that place varus or valgus stresses on the collateral ligaments should be avoided.

Arthrodesis may be indicated for shearing injuries if the degree of tissue loss, especially articular cartilage, makes joint reconstruction impossible. Hyperextension injuries that result in subluxation or luxation of the tarsometatarsal joint, or the proximal or distal intertarsal joints also require partial arthrodesis for a good outcome.⁴⁰ Arthrodesis is performed by application of internal or external fixation ([Figure 32-32](#)). If internal fixation is used, the distal limb is supported by a splint or cast for 6 to 8 weeks or until there is radiographic evidence of fusion. Remobilization of adjacent joints, weight-bearing exercises, and muscle reconditioning are begun after the cast has been removed (see [General Physical Rehabilitation Management of Joint Arthrodesis](#)).



Figure 32-32 Pantarsal arthrodesis as a treatment for a severe shearing injury of the hock.

Injuries of the Common Calcaneal (Achilles) Tendon

The common calcaneal (Achilles) tendon consists of three tendons that insert on the tuber calcanei: the gastrocnemius; the common tendon of the biceps femoris, semitendinosus, and gracilis muscles; and the tendon of the superficial digital flexor muscle. Rupture of the common calcaneal tendon can occur at the musculotendinous junction, the midsubstance, near the tendon's insertion on the tuber calcanei, or the tuber calcanei may fracture. The tendon may be injured by sharp penetrating trauma or chronic repetitive use. If the tendon is avulsed, the end may be palpable 2 to 3 cm proximal to the tuber calcanei. The distal end of the tendon becomes enlarged and firm as fibrous tissue develops. Diagnosis is made by physical examination, ultrasound or other imaging, such as radiographs to identify avulsion fractures or other conditions. An avulsed tendon may be sutured to bone tunnels in the calcaneus. A large bone avulsion fragment may be fixed with pin and tension band. In cases of ruptures at the musculocutaneous junction or in mid-tendon, the tendon is usually sutured primarily. Partial tendon ruptures may be managed by a cast or splint.

After tendon surgery, the hock is often immobilized in a normal standing angle to prevent hock flexion and excessive tension on the repair (Figure 32-33). However, the tendon will still be subjected to some tension from muscle contractions.⁹⁴ Therefore placing a walking ring under the foot and incorporating this into a transarticular external



Figure 32-33 A transarticular external fixator is sometimes used to stabilize a repaired common calcaneal tendon.

fixator or cast will allow weight-bearing forces to bypass the healing tendon. Three or more weeks of immobilization with a splint, cast, positional screw, or external fixator is typical. The walking ring may then be gradually raised, so that partial weight bearing is shared between the ring and foot, until complete weight bearing by the foot occurs. Some studies question which immobilization technique is superior.^{94,95} Rigid fixation should be used for the shortest time necessary. Early partial weight bearing and joint movement create stress on the reconstructed tissue to promote parallel collagen alignment in the repaired tendon and increased early tendon strength.^{96,97} External skeletal fixators may be applied with a hinge to allow limited motion. External fixators may also provide progressive loading of the tendon by performing a staged removal.⁹⁷ After 3 weeks controlled motion is recommended to promote remodeling of the peritendinous scar, which allows the tendon to glide. By 6 weeks the tendon has not attained full strength but should be strong enough to support partial weight.⁹¹ The tendon may continue to slowly increase in strength for a year or more. During the remodeling phase therapy is directed at maintaining tissue mobility, preventing adhesions, and improving proprioception and muscle strength.

Pulsed, 3-MHz therapeutic ultrasound may be used to stimulate collagen repair.⁹⁸ Therapeutic laser also may be beneficial. After the period of immobilization, continuous ultrasound or hot packs may be used to warm the tissues before stretching.⁹⁹ Tendon strength and joint flexion may improve over several months to a year. During this time explosive motion that places large tensile forces on the tendon (e.g., jumping) should be avoided. The long-term prognosis is good.

Luxation of the Superficial Digital Flexor Tendon

The tendon of the superficial digital flexor muscle may luxate medially or laterally from its location as it crosses the calcaneus. It is a traumatic injury, but dysplasia of the tuber calcanei may be a predisposing factor in some cases.¹⁰⁰ Luxation of the tendon may be palpated while the hock is flexed and extended. Treatment is surgical repair of the torn retinaculum. The healing tissues are protected by immobilization of the hock in a splint for 2 to 3 weeks. After the splint has been removed, ROM and weight-bearing exercises are begun. Tissue mobilization may be applied to the area to prevent fibrosis of the tendon, which could limit motion of the digits. The prognosis is good if the repair is done early. Chronic tendinitis and bursitis may decrease the prognosis.

Luxation of the Metatarsophalangeal and Phalangeal Joints

These conditions are the same as described for the front feet.

Table 32-6 Rehabilitation of Joints with Osteoarthritis

Goals	Treatment
Weight management	Dietary counseling and exercise program to achieve and maintain lean body weight.
Pain management	Nonsteroidal antiinflammatories $\pm$ other analgesics as needed $\pm$ Transcutaneous electrical nerve stimulation $\pm$ Laser $\pm$ Shockwave therapy Hot or cold pack of joint
Strengthen periarticular muscles	Hot pack, therapeutic ultrasound, or massage for muscle spasms Neuromuscular electrical stimulation if too weak or painful for active exercise Controlled, low-impact exercise (walking or aquatic exercise). Ongoing patient assessments of lameness and pain are used to continually adjust the intensity of the exercise program.
Limit excessive joint stress	Lifestyle changes, such as ramps instead of steps or jumping Encourage controlled, low-impact activities.
Maintain or improve joint range of motion and proprioception	Passive range of motion and/or joint mobilization Active range of motion—therapeutic exercise (walking or aquatic)
Maintain or improve cartilage health	Passive range of motion Weight-bearing exercise with low-impact (walking or aquatic) Disease modifying osteoarthritic agents
Limit inflammation	Nonsteroidal antiinflammatories to treat chronic or acute inflammation Cryotherapy for episodes of acute inflammation Limit high-impact or uncontrolled activities Avoid obesity Diet containing omega-3 fatty acids, or fish oil supplement

Trauma of the Superficial and Deep Digital Flexor Tendons

These conditions are the same as described for the front feet.

General Physical Rehabilitation Management of Osteoarthritis

Osteoarthritis may also be termed *osteoarthrosis*, *degenerative joint disease* (DJD), or *spondylosis* if in the spine. It is a deterioration of joint health that usually occurs secondary to joint incongruity, instability, or some other disruption of the articular cartilage. Mechanical and biochemical changes that occur within the joint result in decreased cartilage resiliency, cartilage thinning, subchondral sclerosis, synovitis, and osteophyte formation. Clinical signs include joint pain, crepitus, and stiffness. Normal ROM may be lost because of joint surface incongruity, muscle spasm and contracture, periarticular fibrosis, or mechanical block from osteophytes or joint mice. The joint(s) may be enlarged because of synovitis, synovial effusion, osteophytes, or periarticular fibrosis. The pain and stiffness may lead to lameness or decreased activity, with a resultant loss of muscle mass and strength.

The goals of treating OA are to manage pain, maintain function and ROM, and maintain or regain normal activity (Table 32-6). These goals are primarily accomplished by means of weight management, therapeutic exercise, physical modalities, and medications.¹⁰¹⁻¹⁰³ Weight reduction is essential for obese animals, and a lean body weight is maintained by counseling the owners on proper diet and exercise. Weight reduction alone can cause a significant improvement in clinical signs in dogs with hip dysplasia that are more than 10% above their ideal body weight.¹⁰² Excess weight can cause increased mechanical stresses on load-bearing joints. In addition, body fat has metabolic activity that promotes inflammatory reactions within the body, which may contribute to the pathophysiology of OA.

A controlled, low-impact exercise program alone improves pain and overall function scores in geriatric dogs with OA.¹⁰³ Regularly performed, low-impact exercises such as leash walking, walking in water, or swimming, help to maintain supporting muscle strength and joint function while minimizing joint stresses. Bursts of vigorous activity, such as running or jumping, may exacerbate inflammation and should be minimized. Certain activities may be used to target specific muscle groups. For example, repeated sit-to-stand exercises or walking up stairs may

help strengthen the hindlimbs. Exercise sessions should be challenging, but should not be followed by any increase in pain. The exercise program must be tailored to the individual and adjusted to account for the fluctuations in clinical signs that are typically seen with OA.

Other techniques may be used to enhance the exercise program. Use of a walking cart may be indicated in some cases with severe, chronic OA to allow independent function and reduce the risk of falling. In cases of severe muscle loss and weakness, neuromuscular electrical stimulation (NMES) may be used for muscle strengthening. Passive range of motion (PROM) and active range of motion (AROM) exercises are used to improve joint ROM and promote cartilage metabolism and diffusion of nutrients. Swimming can be used to improve ROM, particularly joint flexion.⁸¹ Walking up down hills or stairs may also promote AROM of the joints.¹⁰⁴ Passive stretching alone can improve ROM in dogs with OA.¹⁰⁵ Joint mobilization may also help improve joint motion, cartilage health, and manage pain. Joint compressions, joint mobilizations, PROM, and AROM may be used to improve joint proprioception.

Physical modalities may be incorporated into the therapeutic program. Hot packs, therapeutic ultrasound, or massage may be useful to reduce pain associated with muscle spasms and decrease joint stiffness. These modalities can also be used to increase blood supply to the muscle in preparation for an exercise session. Transcutaneous electrical nerve stimulation (TENS) has been shown to be of benefit in dogs with osteoarthritic pain.⁷⁸ Laser therapy has been used to treat OA, although there is little evidence to support its effectiveness. Extracorporeal shockwave therapy also shows promise as a method to decrease pain in dogs with OA, however, the number of cases reported is limited.¹⁰⁶⁻¹⁰⁹ Cryotherapy can be used to reduce acute inflammatory episodes associated with overuse or postexercise inflammation.

A number of drugs and other substances are promoted for the treatment of OA. Nonsteroidal antiinflammatory drugs may be used as needed for pain management and to treat chronic inflammation or episodes of acute inflammation. It is advisable to maintain the lowest effective dose, to minimize the risks of adverse side effects. The efficacy of NSAIDs varies between individuals, so if the animal is not responding well to one NSAID another may be more effective.¹¹⁰ It is advisable to stop one NSAID completely before starting another because of the potential for adverse side effects. Additional analgesics such as gabapentin or amantadine may be needed in severely affected animals. Although corticosteroids have an excellent antiinflammatory effect, they also promote degenerative changes in the joint and should be avoided for chronic use.¹¹¹

Slow-acting disease-modifying osteoarthritic agents (DMOAs), or chondroprotective agents, have the potential to improve joint health by stimulating cartilage metabolism,

stimulating hyaluronan synthesis, inhibiting periarticular fibrin formation, and inhibiting catabolic enzymes. Some of these agents may also function as antiinflammatory agents or free radical scavengers. Injectable hyaluronic acid or a polysulfated glycosaminoglycan (Adequan) may be of benefit for some arthritic animals.^{112,113} Several oral formulations of DMOAs have been developed. Oral glucosamine hydrochloride, chondroitin sulfate, avocado-soybean unsaponifiables, and manganese ascorbate may have a synergistic effect in dogs to reduce inflammation and lameness.¹¹⁴ S-adenosylmethionine (SAME) appears to be effective in humans with OA, but effectiveness in dogs has not been confirmed.¹¹⁵ The oral forms of DMOAs, known as nutraceuticals, are not regulated as drugs, so there may be considerable variation in product content and quality. There are a number of compounds that are marketed as being effective in the treatment of OA, but have not been proved to be effective by scientific studies. Although the efficacy of most DMOAs is still unclear, the risk of adverse side effects appears to be low. Supplementation of dog food with fish oil, or feeding a diet that contains increased levels of omega-3 fatty acids can alleviate signs of OA.¹¹⁶⁻¹¹⁸ Adipose-derived stem cell therapy shows some promise as an adjunctive treatment to improve the clinical signs of OA.¹¹⁹

Lifestyle modifications may be beneficial for some animals with OA. They should be housed in a warm, dry area with good footing and provided with a padded area for sleeping. Stair climbing and jumping may be minimized by the use of ramps.

The animal should be assessed routinely to alter the rehabilitation program as needed. Functional assessments should be made. Radiographs are of minimal use because they correlate poorly with limb function.¹²⁰

Joint replacement surgery is performed in a few major canine joints. The usual indication is to treat severe OA that cannot be adequately managed medically. Total elbow and stifle replacement surgery is still in its infancy and is not commonly performed.^{41,73} By contrast, total hip replacement is commonly performed in surgical practices for coxofemoral OA, articular fractures, or hip luxation. The complication rate for total hip replacement is reported to be as high as 17%, depending on the system being used and the experience of the surgeon.^{121,122} However, most dogs have a successful final outcome.

General Physical Rehabilitation Management of Excision Arthroplasty

Excision arthroplasty involves the surgical removal of part of the joint, allowing a pseudoarthrosis (false joint) to form from fibrous tissue. This technique is generally limited to the shoulder and hip joints. It is done as a treatment for severe OA, irreparable fractures involving the joint, severe or recurrent joint luxation, or congenital joint deformities.

Elimination of the joint and the bony contact relieves the joint pain. The goal is for fibrous tissue to create a pseudoarthrosis with no bony contact, which provides pain-free function. However, normal function and gait are not expected due to the biomechanical changes. After excision arthroplasty early active use of the limb is encouraged to prevent excessive fibrosis and loss of motion of the pseudoarthrosis. Adequate analgesia throughout the rehabilitation program is a key factor for a successful outcome. Opioids may need to be combined with NSAIDs for pain control. Cryotherapy can also help reduce pain and early inflammation. Passive range of motion exercises are begun during anesthetic recovery and continued daily until the animal is using the limb well. Assisted weight bearing activity is initiated the day after surgery. Once the acute inflammation has subsided, hot packs or therapeutic ultrasound are used to facilitate PROM and stretching. Massage may also be helpful before passive or active exercises. In some cases weight shift exercises are used in the early rehabilitation period to encourage weight bearing. Active weight-bearing activities are progressed to higher levels as tolerated by the animal, to improve limb use and muscle strength.

Femoral head and neck ostectomy (FHO) is the most common excision arthroplasty (see [Figure 32-20, B](#)). After FHO the femur tends to be located more dorsally than normal so there is a functional shortening of the limb, which can cause gait abnormalities. In addition the hindquarters may appear to be asymmetric. Rehabilitation after FHO follows the same guidelines as for any excision arthroplasty, but there is an emphasis on regaining hip extension ([Table 32-7](#)). In addition to PROM for hip extension, ground treadmill or underwater treadmill exercises may be used to promote active hip motion. Sit-to-stand exercises may be used soon after surgery to strengthen hindlimb muscles, without encountering the pain associated with full hip extension. Walking on an inclined

treadmill may promote active hip motion, although the inclination may need to be greater than 5% to induce much increase in hip ROM.⁵¹ Walking up hills or steps can be initiated to promote hip extension¹⁰⁴ and improve muscle strength after the animal is ambulating well. Dancing exercises encourage muscle strengthening and hip extension, especially dancing backward. Although swimming is a good conditioning exercise, it does not promote hip extension.⁸¹ Walking in an underwater treadmill may promote more functional activity if the water level is at the stifle or lower. Cryotherapy may be used at the end of each therapy session to reduce inflammation. Most animals will toe-touch consistently within 1 to 2 weeks, partially weight bear in 3 weeks, and be actively using the leg by 4 weeks. The animal should regain near-normal walking and trotting gaits, but full ROM is rarely achieved, with hip extension being the most limited. Prognosis for return to daily function is generally good but varies with the surgical technique and the chronicity of the preexisting lameness.^{123,124} Large animals tend to have more difficulty with recovery than small animals, and obese animals tend to have more difficulty than athletic animals. Adherence to a rehabilitation program may also affect the outcome, so it is important to manage pain during the rehabilitation period to facilitate animal compliance.

General Physical Rehabilitation Management of Joint Arthrodesis

Arthrodesis is the surgical fusion of a joint that is done to salvage limb function when there is severe joint damage or in certain cases of nerve damage to a distal limb. To create bony fusion the joint must be rigidly immobilized after removing the articular cartilage and placing a bone graft. The joint is usually fused in a functional standing angle (see [Figures 32-13](#) and [32-32](#)). Goniometric measurement

Table 32-7 Rehabilitation After Femoral Head and Neck Ostectomy

Goals	Treatment
Control inflammation	Cryotherapy immediately after surgery and then daily until inflammation resolves, particularly after exercise
Maintain range of motion	Passive range of motion from immediately postoperatively until using the limb well. After the inflammation of surgery has subsided, daily hot packs or ultrasound in conjunction with stretching. Particular attention is paid to hip extension and also abduction. Incorporate dancing and incline walking into exercise program after the first week or two.
Control pain	Nonsteroidal antiinflammatories and additional analgesic medication for first the few weeks, as needed. Modalities as needed.
Regain limb function	Encourage weight bearing the day after surgery. Consider neuromuscular electrical stimulation. Begin with controlled slow leash walking and sit to stand exercises. Walk in aquatic treadmill initially to decrease weight bearing on the limb.

of the contralateral joint may be used as a guide if this limb is normal. The joint is most commonly stabilized with a bone plate, but pins, screws, or an external fixator may be used. Arthrodesis of joints that are located more distally on the limb are associated with better limb function and prognosis than arthrodesis of more proximal joints. Arthrodesis is performed most commonly on the carpal and tarsal joints.

Because of the massive inflammation and edema that can result from the surgical procedure, a pressure bandage is maintained during the immediate postoperative period. A cold-compression bandage or continuously circulating cold pressure wrap is preferred. Nonsteroidal antiinflammatory drugs are also used to minimize the inflammatory reaction. The repair is often supported by external coaptation (cast or splint) if internal fixation was used. This rigid support is usually applied in 3 to 5 days, after the edema and swelling have resolved. The external coaptation is usually maintained for 6 to 8 weeks or until there are radiographic signs of fusion. During this time, PROM or joint mobilization may be performed on adjacent joints. Muscle strengthening exercises may be instituted. Although other joints in the limb can compensate to some degree, the gait will never be normal. As the arthrodesis heals, gait training should be performed to restore the gait to the most normal gait pattern possible.

General Physical Rehabilitation Management of External Fixators

Rehabilitation is very important in patients with external fixators. During surgical planning the surgeon should place pins through safe corridors, without crossing large muscle masses, neurovascular bundles, or entering joints. A pin inadvertently placed through a large muscle mass (as can occur with fixation of the femur) or major tendon (e.g., Achilles tendon), may cause enough pain or mechanical difficulty that the patient will not tolerate ROM exercises or be willing to bear weight on the limb. It may be necessary to penetrate some muscle mass, so analgesia and careful attention to pin tract maintenance is important. The frame itself can present a problem in that it may be difficult to apply cryotherapy. Some clinicians apply ice packs to the frame itself, allowing it to act as a thermal conductor.

Passive range of motion, stretching, and massage can be applied to the soft tissues associated with an external fixator. Patients with external fixators may be reluctant to fully flex and extend the joints in the initial postoperative period. Active exercises that encourage limb flexion include walking over cavaletti rails or other objects, and walking on a land treadmill. Fixators are often used because there are open wounds that make internal fixation less desirable, and patients with open wounds should not be exercised in an aquatic environment.

If a fixator must cross a joint, but is not intended to fuse the joint, it is preferable to incorporate hinges in the fixator frame. The hinges can be locked into a rigid position for the animal's daily function. However, the hinges can be unlocked daily to allow the therapist to perform PROM on the immobilized joint.

General Physical Rehabilitation Management of Amputation

Limb amputation may be performed as a treatment for neoplasia or for severe soft tissue, orthopedic, or neurologic disorders. Dogs and cats typically do well after forelimb or hindlimb amputation. In many cases, the animal is already ambulating on three legs as a result of the condition of the affected limb. The other limbs should be thoroughly evaluated for the presence of orthopedic or neurologic conditions. If desired the affected limb may be placed in a sling before surgery to allow functional assessment of the remaining limbs. Some animals can adapt to amputation of more than one limb. They may require the assistance of a custom cart or prostheses, but some animals learn to walk with two legs (Figure 32-34).

The animal should be encouraged to stand on the first postoperative day, but may need supportive assistance. Therapeutic exercise may begin with standing and walking, and progress to more challenging activities that help the animal adapt to the new center of gravity. Uneven terrain or balance boards may be used to improve limb strength and proprioception. Most animals do well after amputation, whether it is a forelimb or hindlimb.¹²⁵ Animals that are overweight or have impaired function of other limbs may have more difficulty adapting to an amputation, especially forelimb amputation.



Figure 32-34 Double amputees can sometimes learn to walk on two legs. However, use of a cart allows more normal body posture, and may help prevent compensatory problems.

Historically limb prostheses have been used rarely in companion animals. This is partly owing to the fact that animals quickly adapt to walking with three legs. It may be argued that a three-legged quadruped will develop compensatory problems, but the average pet-quality animal functions with no obvious clinical impairments. In recent years there has been more demand for animal prostheses to replace the distal extremities. Most prostheses are designed with a sleeve that is custom fit to the residual limb. It is critical that the prosthesis fit appropriately and that keen attention is paid to skin contact sites on a twice-daily basis to help prevent skin breakdown and wounds. It is vital that an experienced animal prosthetist be involved with construction of a prosthetic. It is also usually necessary to remove the prosthesis during periods of rest to help prevent excess stress on tissues. Despite these precautions, wounds over the stump are common and require diligent care. Another method of creating a prosthesis involves surgical implantation of a prosthetic device into the bone of a limb. Bone grows into the porous metal of the prosthesis to form a more stable, osseointegrated prosthesis. There have been a few case reports involving the use of an osseointegrated prosthesis for the tibia.^{126,127} Common complications include infection, which generally necessitates removal of the prosthesis.

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33 Common Conditions and Physical Rehabilitation of the Athletic Patient

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The diagnosis and treatment of common conditions of athletic and sporting dogs are discussed in this chapter. Most of these conditions involve injuries of the soft tissues such as muscles, tendons, or ligaments. Bone and cartilage can also be involved (e.g. subluxations or traumatic fragmentation of the medial coronoid process).

Muscle Strains

Muscle strains are the most frequent injury in human sports and can be minor, with tearing of a few muscle fibers with pain and local spasm, to complete muscle rupture.¹ Acute muscle injuries rarely have been reported in the canine veterinary literature, and discussion of chronic muscle disorders in dogs is limited to a handful of classic syndromes or inflammatory conditions. However, the reports of muscle strains in human sports medicine are bountiful and well documented in terms of evaluation and treatment.²⁻⁴ It seems likely, given the similarities of the human and canine musculoskeletal system, injuries common in the human athlete would be common in their canine athlete counterpart. The low reported prevalence of muscle injury in dogs is probably owing to a failure to diagnose the condition. Inflammation and edema play a large role in muscle strains. Supportive care to treat the inflammation and swelling is often provided; however, in most cases little thought is given to the damage done to the involved muscles and their subsequent rehabilitation. The strength of muscle contraction may be greatly affected by even a minor strain injury. Some muscle fibers heal, but normal microscopic structure is not restored and scar tissue persists.¹ Healing of the muscle by scar tissue predisposes it to reinjury and possibly to muscle contracture or permanent shortening.¹

Regenerative Medicine Therapy for Musculoskeletal Injury

Regenerative medicine therapy for treatment of injuries and defects is not a new concept. However, its application for musculoskeletal injuries has received significant attention in recent years because of its use in several prominent sports figures. In many human sports medicine clinics, regenerative medicine therapies are offered as “off-label”

treatments for injuries involving muscles, tendons and ligaments. Regenerative medicine has been used for several years to treat these types of injuries in horses and more recently in dogs.

Regenerative medicine therapy encompasses several therapies, including platelet-rich plasma (PRP) and stem cell therapy (SCT). These therapies aim to induce healing through self-regeneration rather than scar tissue formation.

Tissue repair is achieved through a complex response predominately mediated by growth factors and cytokines. PRP, an autologous blood concentrate of platelets above the physiological baseline, contains bioactive proteins and several growth factors including:

- Platelet-derived growth factor (PDGF)
- Transforming growth factor-B1 (TGF-B1)
- Transforming growth factor-B2 (TGF-B2)
- Vascular endothelial growth factor (VEGF)
- Basic fibroblastic growth factor (bFGF)
- Epidermal growth factor (EGF)

In addition, PRP contains factors that include antibacterial and fungicidal proteins, metalloproteinases ADP, ATP, calcium ions, histamine, serotonin and dopamine, that may also affect tissue inflammation.⁹

Stem cells, specifically mesenchymal stem cells (MSC) found in bone marrow and adipose tissue, can differentiate into multiple cell lines including bone, cartilage, and fibrous connective tissue.¹⁰ In addition, MSC secrete cytokine and growth factors that reduce inflammation, inhibit programmed cell death of the cells within the tissue, recruit circulating stem cells to the area, and integrate and reform tissue.¹¹

Synergy between stem cells and PRP has been reported.¹²⁻¹⁸ Certain growth factors and cytokines released from platelets bind to receptors on the surface of the stem cell and initiate a cascade involving signal transduction, gene expression and stem cell proliferation. PRP also provides a delivery vehicle or scaffold to support cell survival and proper differentiation.¹⁹

Anecdotal reports of faster recovery, improved function and reduced recurrence have been encouraging, however, there have been relatively few high-level clinical studies

evaluating efficacy. Studies are currently underway to validate use of regenerative medicine therapy for these types of injuries in humans, horses and dogs. Regenerative medicine therapy should be considered for treatment of some conditions in this chapter. Rehabilitative therapy modalities to promote the activity of and prevent damage to the injected cells is paramount to successful treatment.

Forelimb Conditions

Most forelimb injuries sustained by performance dogs involve the shoulder. The shoulder joint is able to withstand large forces and provide extraordinary mobility while maintaining the stability and control necessary to enable precise function of the forelimb during activity. More than 25 muscles are required for shoulder motion, which involves components of flexion, extension, rotation, abduction, and adduction.²⁰ Given the muscular contribution and complexity of this intricate network of interlinked support mechanisms, the potential for muscle injuries causing performance issues and lameness is significant (Figure 33-1). When muscle strains occur during activities, muscle fibers can be disrupted, most commonly near the muscle-tendon junction.²⁰ These injuries are typically characterized by initial inflammation and pain, followed by healing with marked scar tissue, impeding the ability of the tendon to stretch.²¹ Muscle strains have been reported in every major muscle group of the forelimb, but the muscles at particular risk in performance dogs are the biceps brachii and the supraspinatus.

Bicipital Tenosynovitis

One of the most common shoulder conditions seen in performance dogs is bicipital tenosynovitis (BT), which involves the biceps brachii muscle and its tendon that

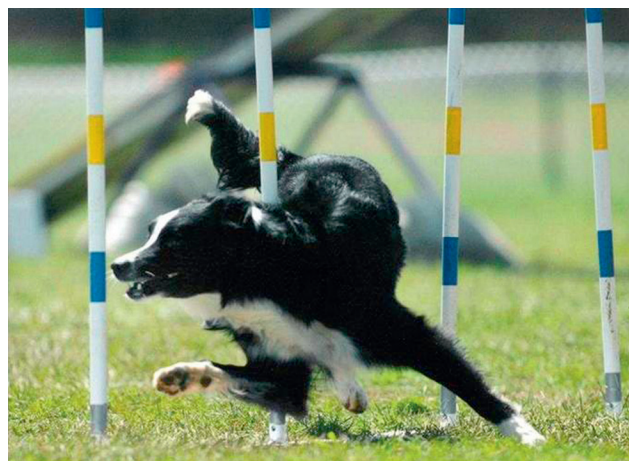


Figure 33-1 Dog going through weaves demonstrating biomechanical forces acting on the shoulder.

crosses the shoulder joint (Figure 33-2). The biceps flexes the elbow and extends and stabilizes the shoulder joint during standing or during the weight-bearing phase of locomotion.²²

Cause

Repeated strain injury appears to be the primary cause of BT in the performance dog. This includes two-on/two-off contacts, landing vertically on the forelimbs from a misjudged jump, overstretching of the muscle, quick turns, and repetitive contractions of the muscle with the shoulder flexed and/or the elbow extended. Injury to the tendon can occur in a number of ways, including strain from overloading, degeneration, or disruption.²³ A single less than maximum load may injure some of the fibers without complete failure of the tendon, but the blood supply to the tendon proper is poor, leading to a longer healing time.²¹ Repetitive strain injury may initiate actual degeneration of the tendon.²¹ As the area continues to be reinjured, the tendon may weaken sufficiently for inflammation or microtears (tendinopathies) in the connective tissue in or around other tendons to form, ultimately leading to shoulder joint instability.²⁴

Diagnosis

Performance dogs with BT commonly have difficulty with quick turning to the affected side and are reluctant to jump. Agility dogs may also have issues performing two-on/two-off contacts and knocking bars with their forelimbs. On gait analysis of dogs with BT often have a shortened stride and a weight-bearing lameness (subtle to severe) that becomes worse with activity. Pain may be elicited by direct

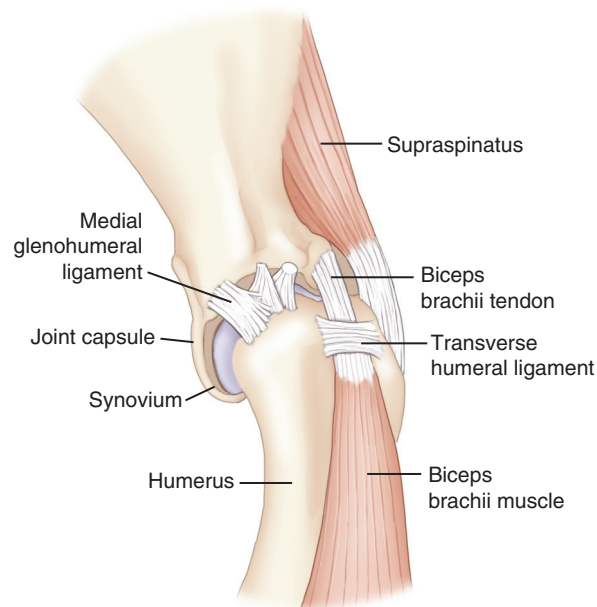


Figure 33-2 Normal canine shoulder anatomy (medial view).



Figure 33-3 Biceps stretch test.

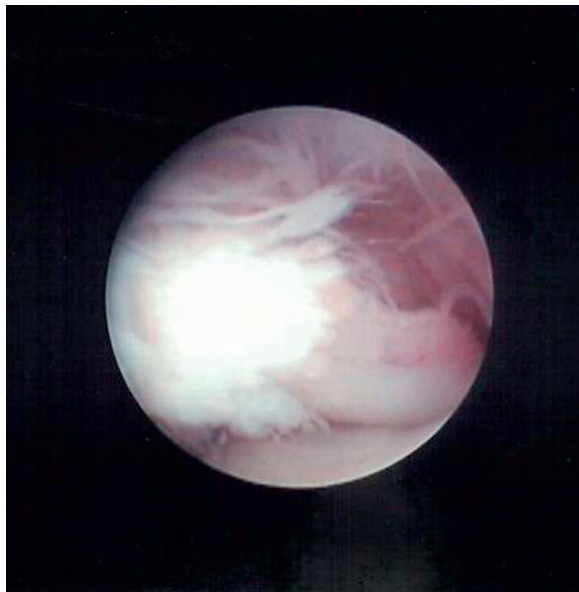


Figure 33-4 Normal arthroscopic image of the canine biceps tendon.

palpation over the biceps tendon. Pain and spasm may be noted when flexing the shoulder while at the same time extending the elbow (**Figure 33-3**).³ Radiographs are of minimal assistance when the injury is in its acute phase, but may reveal mineralization of the tendon when the condition is chronic.²⁴ Magnetic resonance imaging (MRI) and ultrasound may be used to identify the condition in acute and chronic situations.^{23,24} Shoulder arthroscopy is recognized as superior to other diagnostic procedures for diagnosis of bicipital tenosynovitis (**Figures 33-4** and **33-5**).²³

Treatment

Acute cases of bicipital tenosynovitis may be treated by conservative medical management and rehabilitation therapy. Components of conservative medical management

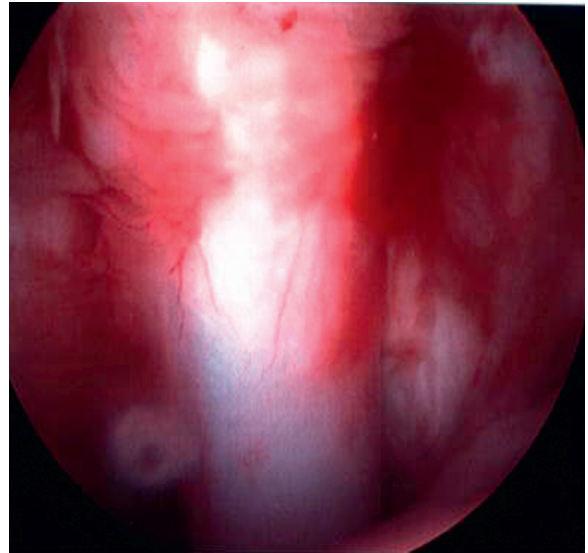


Figure 33-5 Arthroscopic image demonstrating severe bicipital tenosynovitis.



Figure 33-6 Laser therapy of the biceps tendon.

should include controlled activity, nonsteroidal antiinflammatory drugs (NSAIDs), and cryotherapy. Intraarticular injections of hyaluronic acid or steroids, or intra-tendinous injections of platelet rich plasma (PRP), stem cell therapy (SCT) or a combination of these may be considered.^{20,23-25} Regenerative medicine therapy should be considered if a core lesion is present. To promote healing and decrease adhesion formation, laser therapy or pulsed ultrasound may be of use to assist healing to (**Figure 33-6**).²⁰ Initiating low-load, pain-free, high-repetition exercise and range of motion (ROM) movements early in the healing process is important. Pain relief may be achieved with transcutaneous electrical nerve stimulation.²⁰ Acute tendon injuries should not be stretched because of the potential for producing more microtears caused by the chemical changes in the tissue.^{1,20} Pain-free, passive ROM movements (flexion and extension of the limb while the dog is not weight bearing)

and gentle wobble board activities (all directions, front to back and side to side), and alternating leg lifts are recommended first, with progression to active ROM exercises (weight-bearing exercises that cause flexion and extension of the limb). All exercises in the acute stage are aimed at increasing the strength of the shoulder and scapular stabilizers. Balance and proprioceptive exercises are required and are imperative to the healing process. An active dog requires stabilization of this region and strength of the slow twitch muscle fibers before resuming activity. Deep friction massage is used at this stage if it is not too painful.^{25,26} Deep friction massage involves a sweeping motion made across the bicipital tendon (perpendicular to the fibers) for 5 minutes, daily if possible. The goal is to prevent adhesion formation and help realign the fibers of the inflamed tendon. After the cross-friction massage, cryotherapy and laser therapy are recommended before further treatment. A strengthening program consisting of controlled leash walking, stepping over cavaletti rails, wobble board use, and stabilizing exercises should follow. Examples of stabilization exercises include balancing on a well-controlled rocker board, standing on a therapy ball, weight shifting exercises, wheel-barrowing the dog so its weight is on the forelimbs, walking in figure eights, and walking in circles. Leash walks of increasing duration and end-stage eccentric exercises, such as walking and trotting down hills may then be introduced. Short periods of off-lead activity are allowed after leash walks have substantially increased without the return of lameness. In the acute stages, it is imperative to reinforce the owner or handler not to perform any activity that produces lameness. Many owners of performance dogs are eager to return to sports and often push their dogs too quickly.

With chronic BT the inflammatory process must be reinitiated to break down scar tissue and remodel the tendon fibers into the proper orientation, which increases the tendon's load capacity. Because of the need to reinitiate the inflammatory response, NSAIDs or intraarticular corticosteroids are not recommended for chronic tendinopathies. Appropriate therapy for chronic tendinopathies includes deep cross-friction massage, heat, ultrasound therapy, and pain-free ROM/stretching. At this stage cross-friction massage is applied for approximately 10 minutes, two to three times per week. Acupuncture and laser to increase circulation and for pain control may be beneficial as well. The exercise progression is similar to that recommended for acute tendinopathies, but stretching after treatment is advised. Active ROM exercise such as walking for longer periods also is recommended. End-stage eccentric exercises, such as walking and trotting down hills, should be added followed by short periods of off-lead activity provided lameness is completely resolved.

Because of the chronic nature of the tendon changes, the dog usually takes longer to recover and progress through the healing and exercise stages. It is again imperative to

convey to the owner and handler that the progression of conservative management of BT may be lengthy and nothing should be done to aggravate the condition. The dog should be pain free and not exhibit any lameness.

Regenerative medicine therapy may also be considered in the treatment of chronic BT, however, modification of the rehabilitative therapy protocol is necessary to promote healing and prevent damage to the injected cells.

Surgical treatment is recommended for biceps avulsion-tears and chronic cases of BT that are not responsive to medical management or rehabilitation therapy. Surgical options include tenodesis (releasing and reattaching the tendon at a distal location on the proximal humerus) or an arthroscopic tendon release.²⁷ Controlled activity is an important component during the postoperative period and should include leash walks of increasing duration and end-stage eccentric exercises, such as walking and trotting down hills. Short periods of off-lead activity are allowed after the lameness has resolved. Nonsteroidal antiinflammatory drugs may be used to decrease pain and inflammation following surgery. Modalities as described for the acute treatment of BT may be used postoperatively to improve comfort and function and facilitate progression to the retraining phase of recovery.

Toward the end of rehabilitation therapy sport-specific retraining should begin. Examples of retraining techniques for BT may include starting with low, straight-line jumps with very wide sweeping turns that progress over time to higher jumps with tighter angles. Do not include the A-frame and weave poles until late in the retraining period. Alterations in contact controls from a two on/two off to running or four on the floor may be necessary. Be sure to start with turns toward the injured limb and progress to turns toward the uninjured limb.

Dogs that will be returning to jumping activities should start with low-level activities before returning to the sport.^{4,26} In jumping activities, the biceps muscle and tendon act eccentrically to control the landing of the body. After cavaletti work has been employed, low-level jumping can be initiated. The jump heights should be set lower than the dog's normal jump height. For example, if the dog normally jumps 22 inches in competition, the jump height should be set at 16 inches. Three or four jumps should be set up and the surface should be padded or the exercise should be performed in sand. Three to five repetitions should be performed of the jump sequence—the dog going over the jumps—and then the dog should be reevaluated. The jump sequencing should not cause any lameness and should only be performed three times per week with at least a 48-hour break in between. Plyometrics should be progressed by increasing the jump height and the number of repetitions. Once the dog is up to its normal jump height on a controlled surface, additional training surfaces can be added.

Following a warm-up, make sure to perform active stretching techniques. Following agility work, use an

appropriate cool down with cryotherapy and passive stretching techniques (including shoulder flexion with elbow extension). Treadmill walking and the use of an underwater treadmill are also useful for the rehabilitation of BT. In acute stages exert caution with swimming because the shortened position of the biceps tendon with rapid joint motions may aggravate the condition.

Supraspinatus Tendinopathy

In humans and dogs several degenerative disorders of the supraspinatus tendon have been identified, including rotator cuff tears, calcifying tendinitis or tendinosis (microtears), and tendinosis as a result of overuse.^{28,29} In humans degeneration of the supraspinatus tendon is believed to be a contributing factor in the development of rotator cuff tears.²⁹ There is good evidence that overuse is likely an important factor in this disorder. At the cellular level affected tendons contain discontinuous, disorganized fibers, and typically, no acute inflammation is detected.²⁹ In chronic cases a rapidly growing nodule develops that can impinge on the biceps brachii tendon and cause pain.³⁰ The supraspinatus muscle extends the shoulder and advances the limb. The muscle is important to stabilize and prevent collapse of the shoulder joint and is active during 65% to 80% of the time when the dog is standing (Figure 33-7).²²

Cause

The cause of this injury in performance dogs appears to be related to repeated strain injury. This includes hitting the ground or agility contacts on an outstretched forelimb, slipping, overstretching, or overuse of the muscle, quick

turns and repetitive eccentric contractions (contractions during lengthening of the muscle), as well as concentric contractions (contractions with shortening of the muscle) with the muscle in a lengthened state.

Diagnosis

Performance dogs with supraspinatus tendinopathy (ST) commonly have weight-bearing lameness on one side that becomes worse with activity and often is resistant to treatment. The supraspinatus muscle may be atrophied and direct palpation over the tendon and flexion of the shoulder may cause pain (Figure 33-8). Concurrent BT or shoulder instability may be present, so a thorough shoulder examination is required. Magnetic resonance imaging readily demonstrates the condition in the acute phase (Figure 33-9). Plain radiographs and computed axial tomography (CT), which reveals a cross-section of an internal body structure) may reveal mineralization in chronic cases. Musculoskeletal ultrasound is this author's preferred diagnostic tool for this condition. This technique provides a non-invasive definitive diagnosis of ST and allows for serial evaluations to assess response to treatment. Arthroscopic exploration is recommended and may demonstrate impingement of the biceps tendon secondary to supraspinatus tendon swelling as well as concurrent pathologies, including medial shoulder instability in severe cases.

Treatment

Controlled activity, retraining, and preventive techniques for acute and chronic ST is similar to that used for bicipital tenosynovitis. Similar to the rehabilitation of BT, rehabilitation should be initially aimed at reducing the inflammation and increasing the strength of the shoulder and scapular stabilizers.⁴ After the inflammation has decreased and the core strength of the shoulder complex has improved, more dynamic activities may be initiated.

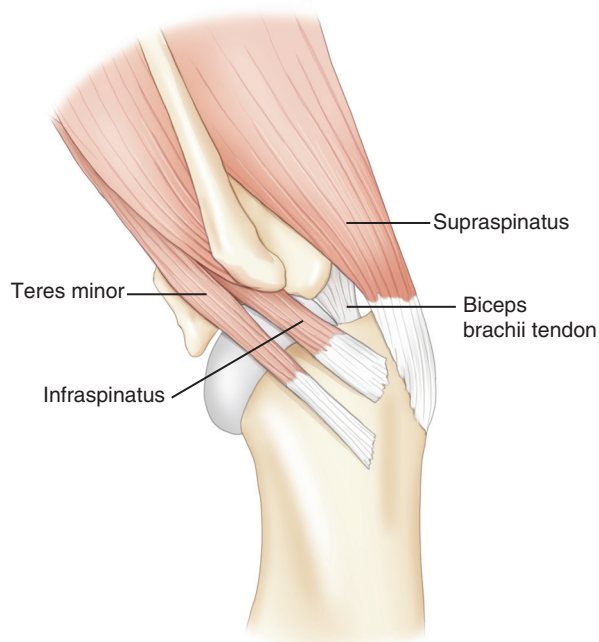


Figure 33-7 Normal canine shoulder anatomy (lateral view).



Figure 33-8 Palpation of the supraspinatus tendon point of insertion.

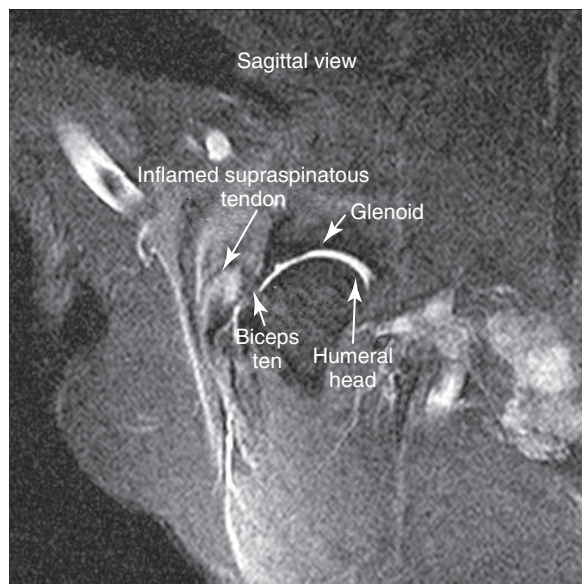


Figure 33-9 Magnetic resonance image of a supraspinatus tendinopathy.

Treatment for acute cases consists of conservative medical management (controlled activity and NSAIDs) and rehabilitation therapy (manual therapy, modalities, ROM with progression to stretching and strengthening exercises), and acupuncture as described for BT.²⁰ Cross-friction massage may be performed to the supraspinatus tendon in the acute phases for 5 minutes daily and 10 to 15 minutes in chronic cases.

As with BT the inflammatory process should be reinitiated in chronic cases and follow a similar rehabilitation protocol. Shockwave and ultrasound therapy have also been described as valuable modalities for the treatment of chronic calcified supraspinatus tendinopathies.³¹

Regenerative medicine therapy should be considered, especially in cases in which a core lesion has been identified. Under ultrasound guidance, cells can be directly injected into the affected tendon. In more severe cases a second injection may be needed. This therapy can be used as the primary treatment of ST, however, arthroscopic evaluation of the shoulder is recommended as concurrent pathologies of the joint are common. In these cases, arthroscopic treatment of concurrent pathologies can be combined with intraarticular regenerative medicine therapy to promote healing and speed recovery. Appropriate modification of the rehabilitative therapy protocol is necessary to prevent damage to the injected cells.

Results of a pilot study of 185 dogs with a confirmed diagnosis of supraspinatus tendinopathy (ST) by ultrasound or MRI performed by one of the authors (SC) provides support of regenerative medicine therapy for treatment of ST. Performance and sporting dogs accounted for just over one third (67) of the population, while companion animals accounted for the remaining two thirds (118). Of the performance/sporting dog population, 48% participated in agility. At presentation, a majority of dogs

had a history of failed response (unresolved discomfort or lameness) to rest and NSAID therapy (134). Seventy-five failed to respond to rehabilitation therapy and 6 received intra-articular injections (hyaluronic acid – 2; prednisolone sodium succinate – 4), which also did not resolve lameness. A unilateral forelimb lameness presentation was more common than bilateral (124 and 61, respectively). Lameness was graded from 0 (sound) to 6 (non-weight bearing lame). The average lameness score was 2.9 (median 3). Duration of lameness ranged from one week to greater than a year. 29.7% of dogs had a chronic (>1 year) lameness. Of those with lameness less than one year, the average duration was 15.2 weeks (median 13 weeks). Pain on direct palpation of the supraspinatus was found in 59% of dogs. Pain or spasm on shoulder extension and abduction was noted in 63.1% and 71.3%, respectively. Pain or spasm on flexion was recorded in 74.6% of dogs and pain on biceps stretch (shoulder flexion with elbow extension) was recorded in 47.5% of dogs. Shoulder radiographs were performed on 161 dogs. Of these, mineralization was noted in 35 cases. Of dogs diagnosed with bilateral disease on presentation (61), mineralization was noted in 8 cases. MRI of the shoulder was performed in 30 cases. Findings indicative of a supraspinatus tendinopathy on MRI include hyperintensity of signal on T1 and STIR sequences of the ST at its insertion on the greater tubercle. Diagnostic ultrasound was performed in 140 cases. In most cases the supraspinatus appeared enlarged (44.3%) with irregular fiber pattern (46.4%) and mixed echogenicity (80.7%). Shoulder arthroscopy was performed in 131 cases. A supraspinatus bulge was noted in 90.8% of dogs. Of these, impingement of the biceps tendon was noted in 59.5% of dogs. While 43.5% of dogs had bicipital tenosynovitis, only 13% had actual pathology of the biceps tendon. Pathology of the subscapularis was observed in 91.6% of dogs. MGL pathology was present in 51.1% of dogs. A tear of the labrum was noted in 29.8%, cartilage defects were noted in 32.8% and bone lesions in 8.3% of dogs. Seventy-four cases were treated with regenerative medicine therapy. Final evaluations were performed on 64 dogs. Of these, lameness resolved in 90% of cases and improvement was noted in the remaining 10%.

Surgical treatment is warranted for chronic cases that do not respond to conservative medical management and rehabilitation therapy. Surgical treatment should include arthroscopic exploration to identify and treat possible concurrent BT or shoulder instability, or removal of abnormal tissue through an open approach.³⁰ Surgery has been considered the treatment of last resort in humans with tendinosis because of the small reported difference in treatment success between surgical intervention and conservative management. Rehabilitation therapy, as described for BT, following surgical intervention is essential in returning the canine athlete to performance.

The retraining protocol for this condition mirrors that of retraining following rehabilitation of BT.

Infraspinatus Contracture

Contracture of the infraspinatus muscle is a rare condition most often affecting large breed dogs, particularly working dogs (i.e., hunting dogs, herding dogs) and other very active dogs.³² Infraspinatus muscle contracture was first reported in the Dutch literature in 1970³³ and is one of the two most frequently reported contractures in the dog, along with quadriceps contracture.^{34,35} Unilateral contraction of the infraspinatus muscle is most common; however, bilateral contractures have also been reported.^{33,36,37} Most cases are working or hunting dogs that present with an acute onset of shoulder lameness with work or exercise.^{33,34,36-40}

The infraspinatus lies within the infraspinous fossa and extends caudally somewhat beyond the fossa. It arises from the fossa, the scapular spine, and the caudal border of the scapula, and at the shoulder joint becomes a strong tendon that crosses the caudal part of the greater tubercle. The infraspinatus is the outward rotator and abductor of the humerus and a flexor or extensor of the shoulder joint depending on the position of the joint when the muscle contracts.

Cause

Causes reported include repetitive microtrauma, blunt trauma, and osteofascial compartment syndrome.^{41,42} Compartment syndrome is a disease process seen in people; however, it can also occur in dogs and horses.⁴³ The pathology to muscle tissue is caused by an increase in pressure in a confined, inelastic space that leads to muscle inflammation with no room to expand.⁴³ Muscle necrosis occurs when the pressure rises to greater than 30 mm Hg (normal, -2 to 8 mm Hg) in a single compartment with no relief.⁴³ A compartmental pressure greater than 120 mm Hg may lead to complete and irreversible nerve block.⁴⁴ Causes for an increased pressure within a compartment include hemorrhage, postischemic tissue swelling, external pressure (bandage), or intramuscular injection.³² In elastic barriers that predispose to compartment syndrome there are structures such as fascia and bone.⁴³ Four different compartments have been described in the dog: femoral, craniolateral crus, caudal crus, and caudal antebrachial.^{43,44}

It may be possible that compartment syndrome may play a role in the contracture of the muscles of the shoulder joint. If an injury is sustained to the infraspinatus muscle that leads to moderate cellular inflammation, the muscle begins to swell and has no room to expand. Inflammation within the boundaries of the deltoids, scapula, and surrounding fascia increase the compartmental pressure and inhibit venous blood flow, initially causing congestion. Then arterial blood flow and innervation are affected, as well as diminished cellular response to damaged tissue. It is the unforgiving pressure that leads to further cellular damage. Cellular swelling caused by pressures rising to greater than 30 mm Hg is what seems to lead to necrosis and/or fibrosis of the affected muscles. Early treatment is focused on surgical decompression by fasciotomy;

however, if contraction has already occurred, tenectomy or myectomy may be necessary.⁴⁵

Electrophysiologic studies of affected dogs indicate that infraspinatus contracture is a primary muscle disorder rather than a neuropathy. Histologically, affected tissues demonstrate degeneration and atrophy of skeletal muscle with fibrous tissue replacement. These findings support the hypothesis that acute traumatic events caused incomplete rupture of the infraspinatus muscle, leading to fibrosis and contracture.

Diagnosis

After an acute onset of a painful non-weight-bearing lameness, the initial pain and lameness improve over a period of 1 to 4 weeks, after which a characteristic circumducted gait abnormality develops in the forelimb. The initial injury to the infraspinatus muscle is not fully recognized or correctly interpreted in most cases, at least not with regard to its potential as a precursor of myopathy. Affected dogs most often present with mature contractures and adhesions of the tendon of insertion of the infraspinatus muscle to the adjacent joint capsule. As a consequence most or all of the internal rotation of the humerus in relation to the scapula is lost. This lack of rotation is a reliable clinical sign for diagnosis of the disorder. The humerus may also be held in slight abduction and external rotation.

Acutely the shoulder is swollen and very painful. The initial trauma usually resolves with rest and supportive treatment. However, the affected limb begins to show signs of contracture several days to weeks after the initial trauma.⁴⁰ Physical examination of a chronic case usually indicates a bright, alert dog, without signs of any systemic illness. There is usually a weight-bearing lameness of the affected forelimb and a characteristic gait. Palpation and manipulation of the contracted limb does not usually elicit a pain response.^{38,40} The remaining shoulder muscles, mainly the supraspinatus⁴⁰ and deltoid muscles, may show signs of disuse atrophy or be contracted as well.^{33,34,39,40} A substantial decrease in range of motion of the shoulder joint is appreciated.⁴⁰ The affected forelimb is usually held with the shoulder in abduction and the elbow in adduction against the thoracic wall³⁸ (Figure 33-10). The distal forelimb and carpus are externally rotated and held in abduction.^{36,40} Gait abnormalities usually consist of weight-bearing lameness and circumduction^{33,38-40} of the affected forelimb with a characteristic fliplike action of the paw when placing the foot.^{36,40} When the patient is placed in lateral recumbency with the affected limb up, a true infraspinatus contracture will cause the distal forelimb to remain in an abducted and externally rotated position off the floor and away from the body.

Siems et al. reported a series of ultrasonographic findings of two concurrently traumatized infraspinatus muscles in the same dog; one resolved without complications, and the other resulted in contracture.³⁹ The primary ultrasonographic description of a contracted infraspinatus muscle is increased echogenicity when compared with a normal



Figure 33-10 Characteristic posture of a dog with contracture of the infraspinatus muscle.

supraspinatus muscle.^{36,39} Magnetic resonance imaging is another diagnostic modality that may be useful for diagnosing the condition in both the acute and chronic phases.

Treatment

Physical rehabilitation of acute or subacute infraspinatus tendinopathies may be used to maintain joint motion and prevent contracture. Continuous ultrasound and stretching exercises may help maintain the length and mobility of the affected tissues. Mature contractures most likely require surgical release. Mature contractures are treated with a tenotomy of the tendon of insertion of the infraspinatus muscle and release of capsular adhesions of the tendon. Infraspinatus muscle contracture carries an excellent prognosis with tenectomy of the infraspinatus tendon.^{40,46} Lateral or craniolateral approaches to the shoulder both provide good exposure of the infraspinatus tendon.^{33,40,41} Complete transection of the infraspinatus tendon with removal of a length of the tendon is required to restore motion in the affected limb and most often proves curative with an immediate release of contracture and return to function of the shoulder joint.^{36,40} It is common to find adhesions to the infraspinatus tendon and joint capsule during surgery; however, after they are dissected away they seldom cause further problems.^{40,41} Postoperative bandages need not be applied, and rehabilitation therapy should start when sutures are removed to prevent the formation of further adhesions.⁴⁰

The postoperative management after adhesion release includes full weight bearing and gentle, pain-free passive ROM, massage, and stretching exercises of the shoulder, elbow, and carpus two to three times daily to maintain ROM and promote normal alignment of the healing tissues. Cryotherapy may be used after exercise. Excessive activity such as unrestricted running or playing should be avoided in the first several weeks to avoid tissue damage and recurrence of fibrous tissue. If the contracture has been

present for more than a few weeks before the surgery is performed, atrophy of other forelimb muscles may be present. General conditioning exercises for the limb, such as progressive walking, wheel-barrowing, and aquatic therapy, should be used to gradually return these muscles to normal size and strength. The use of neuromuscular electrical stimulation may also be considered if the atrophy is severe and the dog is having difficulty using the limb because of weakness.

Medial Shoulder Syndrome and Medial Shoulder Instability

Medial shoulder syndrome (MSS) is one of the most common causes of forelimb gait related issues and lameness in performance dogs. This condition may be considered somewhat similar to rotator cuff injuries in people (including ligament disruption; tendinopathy; labral and capsular tears or disruption). In human athletes, an overhead throwing motion involves intricate motion combined with proper stabilization. During this activity, the shoulder must be lax enough to allow extremes of motion but secure enough to provide stabilization.^{4,26} The same is true for the canine athlete. Many routine maneuvers performed by dogs also involve extremes of ROM and require appropriate stabilization. In dogs with medial shoulder instability (MSI), lameness may be as subtle as performance-related problems such as refusing tight turns or as severe as a weight-bearing lameness. MSI involves multiple components of the shoulder joint. Definitive diagnosis requires a thorough orthopedic evaluation and advanced diagnostic tests. Medial shoulder instability is diagnosed when significant shoulder laxity is present. Early identification and treatment offer the best long-term result and prevent further shoulder instability and progression of osteoarthritis.

Anatomy

Anatomically, the shoulder joint is one of the least stable joints in the dog, relying on soft tissue structures for stability and function (Figure 33-11).⁴⁶ In dogs the glenohumeral (shoulder) joint has the widest ROM and is capable of movements in any direction, but its primary movements during ambulation are flexion and extension.⁴⁶ The glenohumeral joint relies heavily on capsuloligamentous support to provide stability because its anatomic conformation does not provide a stable socket.⁴⁶ The humeral head and glenoid cavity of the scapula make up the ball and socket of the shoulder joint. The glenoid provides little coverage of the humeral head. The joint capsule, ligaments, and surrounding muscles and tendons all contribute to the stability of the shoulder. Insult or injury to any portion of the capsuloligamentous support structures can potentially cause shoulder joint pathology resulting in pain and dysfunction.⁴⁷ The components that are most commonly affected by MSS and MSI include the craniomedial joint capsule, medial glenohumeral ligament, subscapularis